

Rubidium-82 (Sr-82/Rb-82) Generator and Associated Accessories

1. Introduction/Purpose

William Beaumont Army Medical Center (WBAMC) is seeking sources capable of providing a Rubidium-82 (Rb) Generator System and associated accessories for cardiac PET imaging. The goal is to identify vendors who can supply a complete, integrated system compatible with current CardioGen-82® by Bracco procedures to ensure continuous cardiac PET imaging services.

2. Required Technical Specifications & Salient Characteristics

Scope of Requirements - Interested vendors must provide a system that meets the following minimum requirements:

- The contractor must provide a complete, FDA-approved Sr/ Rb generator system (generator column containing stannic oxide, self-contained, shielded cart) and all necessary accessories for producing Rb chloride injection for PET myocardial perfusion imaging (MPI).
- System Compatibility: Generator must be fully compatible with the current Bracco Diagnostics CardioGen-82® Infusion System (Model 1701 or current standard). System must be fully compatible with, or equal to, the current CardioGen-82 system by Bracco Diagnostics, enabling rapid, on-demand serial rest/stress studies (30–35 minutes).
- Isotope Production: Generator must utilize a Strontium-82 parent isotope (approx. 25-day half-life) to produce a short-lived (~75-second) Rb-82 daughter isotope.
- Performance: Must allow rapid sequential rest/stress imaging (30–45 minute study time) and provide high-quality cardiac PET images.
- Safety Standards: Must meet FDA-approved specifications for breakthrough levels (adhering to ALARA principles).
- Clinical Efficacy: Must allow for accurate quantification of myocardial blood flow (MBF).
- Essential Software: Must include current proprietary software (e.g., HeartSee) for imaging, data management, and automated Quality Control. Accessories: Must include all required infusion sets, patient administration sets, waste containers, vented caps, and printer paper.
- Generator Life: Generator must provide a functional lifespan ranging from 4 to 12 weeks, with consistent 82Rb yield, requiring regular replacement/reloading due to Sr-82 parent decay.
- High-Volume Capability: The system must allow for multiple consecutive patient studies (10–15+ per day) with rapid generation times.
- Daily QA/Safety: Must feature automated daily eluate testing and Dose Data reporting capabilities.

- Support: Must provide technical support, including 24/7 emergency contact in case of generator malfunction or breakthrough alerts.

3. Included Accessories & Support

The proposal must include the necessary accessories for operation, including but not limited to:

- Patient Administration Sets (42"): 25 sets/box (1 set required per patient)
- Vented Cap for Waste Bottle: 12 caps/box (
- Waste Bottle: 2 bottles 500mL
- Elution Vial Shield: 1 shield/box
- Thermal Printer Paper: 12 rolls/box
- Accessory Pack: 1 complete set of tubing for Infusion System
- Elution Test Vials: 25 vials/box 50 mL
- Saline Tags: 25 tags/box (Compatible with CardioGen-82®).

4. Generator Lifetime & Delivery

- The generator must be available for a typical lifetime of 4 to 12 weeks, with consistent yield throughout its life.
- Vendor must ensure regular, timely replacement or reloading of the generator to avoid operational downtime.